

SESSION 3 LAB EXPERIMENTS

4. Out of Boundary Paths

```
#include <stdio.h>

#define MOD 1000000007

int main()
{
    int m, n, N, i, j;
    int dp[51][51] = {0};
    int next[51][51];
    printf("Enter m, n, N, i, j: ");
    scanf("%d %d %d %d %d", &m, &n, &N, &i, &j);
    dp[i][j] = 1;
    long long ans = 0;
    for (int step = 1; step <= N; step++)
    {
        for (int x = 0; x < m; x++)
            for (int y = 0; y < n; y++)
                next[x][y] = 0;
        for (int x = 0; x < m; x++)
        {
            for (int y = 0; y < n; y++)
            {
                if (dp[x][y] == 0)
                    continue;
                int ways = dp[x][y];
                if (x == 0) ans = (ans + ways) % MOD;
```

```

        else next[x - 1][y] =
            (next[x - 1][y] + ways) % MOD;
        if (x == m - 1) ans = (ans + ways) % MOD;
        else next[x + 1][y] =
            (next[x + 1][y] + ways) % MOD;
        if (y == 0) ans = (ans + ways) % MOD;
        else next[x][y - 1] =
            (next[x][y - 1] + ways) % MOD;
        if (y == n - 1) ans = (ans + ways) % MOD;
        else next[x][y + 1] =
            (next[x][y + 1] + ways) % MOD;
    }
}

for (int x = 0; x < m; x++)
    for (int y = 0; y < n; y++)
        dp[x][y] = next[x][y];
}

printf("Number of ways = %lld\n", ans);

return 0;
}

```

main.c	Run ▶	Output
<pre> 1 2 #include <stdio.h> 3 4 #define MOD 1000000007 5 6 int main() 7 { 8 int m, n, N, i, j; 9 int dp[51][51] = {0}; 10 int next[51][51]; 11 12 printf("Enter m, n, N, i, j: "); 13 scanf("%d %d %d %d %d", &m, &n, &N, &i, &j); 14 15 dp[i][j] = 1; 16 17 long long ans = 0; 18 19 for (int step = 1; step <= N; step++) </pre>		<pre> Enter m, n, N, i, j: 2 2 2 0 0 Number of ways = 6 </pre>

5. House Robber

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, nums[100];
```

```
    printf("Enter number of houses: ");
```

```
    scanf("%d", &n);
```

```
    printf("Enter money in each house:\n");
```

```
    for (int i = 0; i < n; i++)
```

```
        scanf("%d", &nums[i]);
```

```
    int prev2 = 0;
```

```
    int prev1 = 0;
```

```
    for (int i = 0; i < n; i++)
```

```
    {
```

```
        int current = prev2 + nums[i];
```

```
        if (prev1 > current)
```

```
            current = prev1;
```

```
        prev2 = prev1;
```

```
        prev1 = current;
```

```
    }
```

```
    printf("Maximum money = %d\n", prev1);
```

```

return 0;
}

```

```

main.c
1
2 #include <stdio.h>
3
4 int main()
5 {
6     int n, nums[100];
7
8     printf("Enter number of houses: ");
9     scanf("%d", &n);
10
11    printf("Enter money in each house:\n");
12    for (int i = 0; i < n; i++)
13        scanf("%d", &nums[i]);
14
15    int prev2 = 0;
16    int prev1 = 0;
17
18    for (int i = 0; i < n; i++)
19

```

Run ▶

Output

```

Enter number of houses: 3
Enter money in each house:
2 3 2
Maximum money = 4

```

6. Climbing Stairs

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter number of steps: ");
```

```
    scanf("%d", &n);
```

```
    if (n <= 0)
```

```
    {
```

```
        printf("Number of ways = 0\n");
```

```
        return 0;
```

```
    }
```

```

if (n == 1)
{
    printf("Number of ways = 1\n");
    return 0;
}

```

```

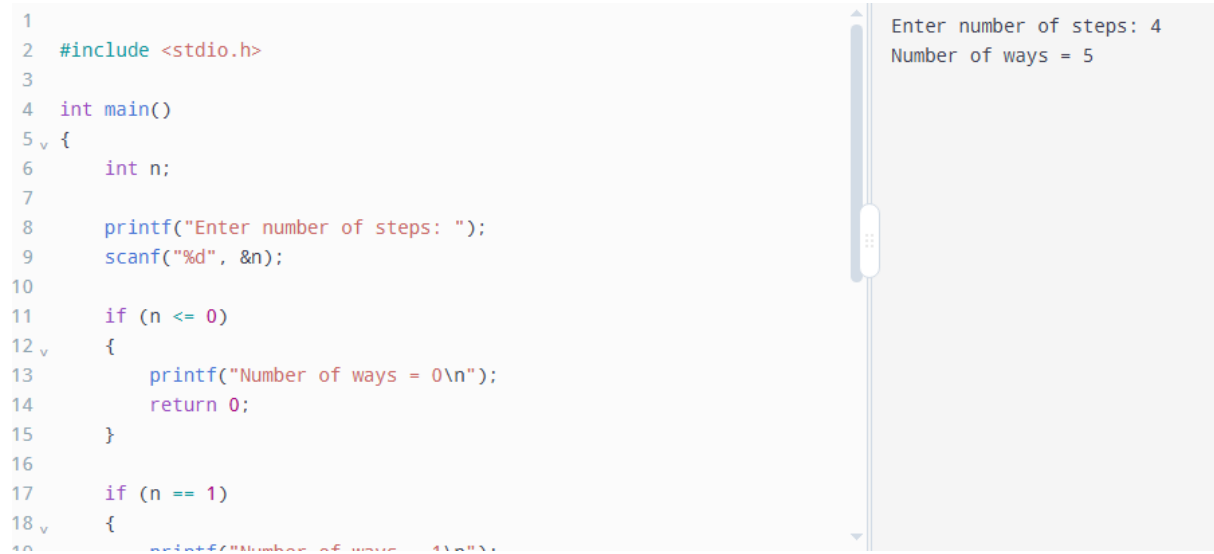
int a = 1, b = 2, c;

```

```

for (int i = 3; i <= n; i++)
{
    c = a + b;
    a = b;
    b = c;
}
printf("Number of ways = %d\n", b);
return 0;
}

```



The screenshot shows a C program in a code editor on the left and its execution output on the right. The code is a recursive function to calculate the number of ways to reach the nth step, with base cases for n=0 and n=1. The output shows the user entering 4 steps, resulting in 5 ways.

```

1
2  #include <stdio.h>
3
4  int main()
5  {
6      int n;
7
8      printf("Enter number of steps: ");
9      scanf("%d", &n);
10
11     if (n <= 0)
12     {
13         printf("Number of ways = 0\n");
14         return 0;
15     }
16
17     if (n == 1)
18     {
19         printf("Number of ways = 1\n");
20         return 0;
21     }
22
23     int a = 1, b = 2, c;
24
25     for (int i = 3; i <= n; i++)
26     {
27         c = a + b;
28         a = b;
29         b = c;
30     }
31
32     printf("Number of ways = %d\n", b);
33     return 0;
34 }

```

Enter number of steps: 4
Number of ways = 5

7. Unique Paths in a Grid

```
#include <stdio.h>

int main()
{
    int m, n;
    long long dp[100][100];

    printf("Enter m and n: ");
    scanf("%d %d", &m, &n);

    for (int i = 0; i < m; i++)
        dp[i][0] = 1;
    for (int j = 0; j < n; j++)
        dp[0][j] = 1;
    for (int i = 1; i < m; i++)
    {
        for (int j = 1; j < n; j++)
        {
            dp[i][j] = dp[i - 1][j] + dp[i][j - 1];
        }
    }

    printf("Number of unique paths = %lld\n", dp[m - 1][n - 1]);

    return 0;
}
```

main.c		Run ▶	Output
<pre>1 2 #include <stdio.h> 3 4 int main() 5 { 6 int m, n; 7 long long dp[100][100]; 8 9 printf("Enter m and n: "); 10 scanf("%d %d", &m, &n); 11 12 for (int i = 0; i < m; i++) 13 dp[i][0] = 1; 14 15 for (int j = 0; j < n; j++) 16 dp[0][j] = 1; 17 18 for (int i = 1; i < m; i++)</pre>			<pre>Enter m and n: 3 4 Number of unique paths = 10</pre>